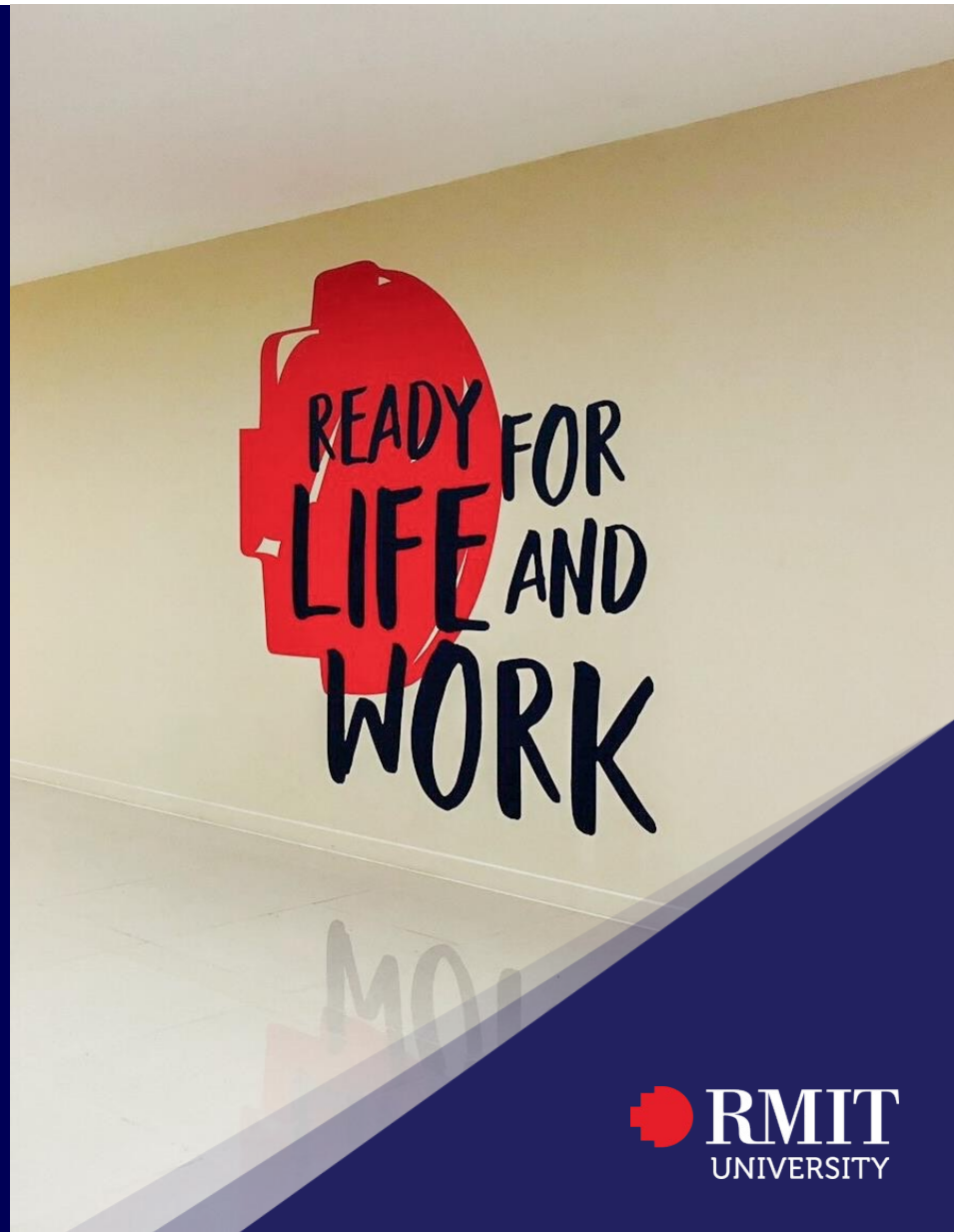


Unsupervised Learning

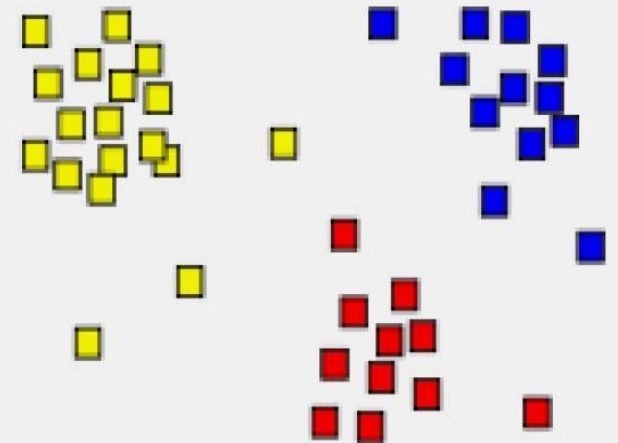
COSC2753 Machine Learning
Semester 1, 2026



Clustering

Clustering

- **Clustering:** finding related groups (called **clusters**) of "similar" instances in a data set.
- There is a lot to learn even in the absence of class labels
 - Searching for typical groups of examples
 - Finding out what is common to examples from group
 - Using these groups to predict unknown (or missing) attribute values
 - What are the most typical features (or combinations of features) to describe the examples



Motion Segmentation



- Identify pixels that belong to each independently moving object in video.
 - Camera may be moving.
- Can be used to predict where the object will be next (or fill in gaps)



Applications



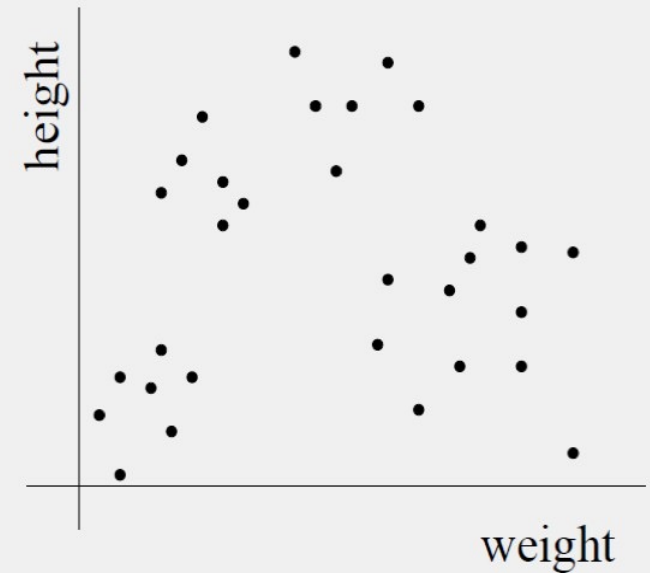
- Recommender systems – recommend new items based on a user's tastes
- Social network analysis – recognise communities within large groups of people
- Anomaly detection – identify outliers with respect to clustering structure in data
- Medical imaging – differentiate between different types of tissue
- Market research – partition the general population of consumers into market segments
- Crime analysis – identify areas that have greater incidences of particular types of crime

What should the clusters be like?



Clusters:

- Should **not overlap**, that is, each instance must belong to one cluster.
- The instances should be “**close/similar**” **within a cluster**
- Should have clear “**differences**” **to other clusters**
- “Distances” between clusters should be maximised

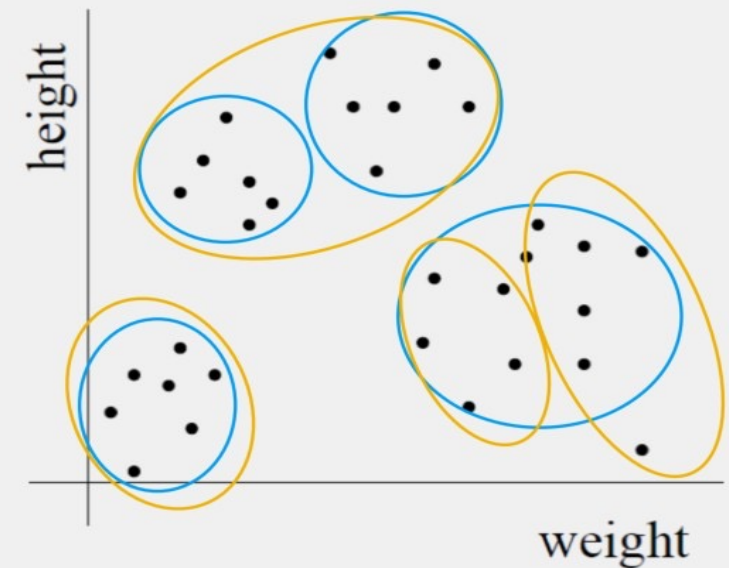


The goal of clustering is to assign each instance
(and unseen data example) to a cluster

What should the clusters be like?

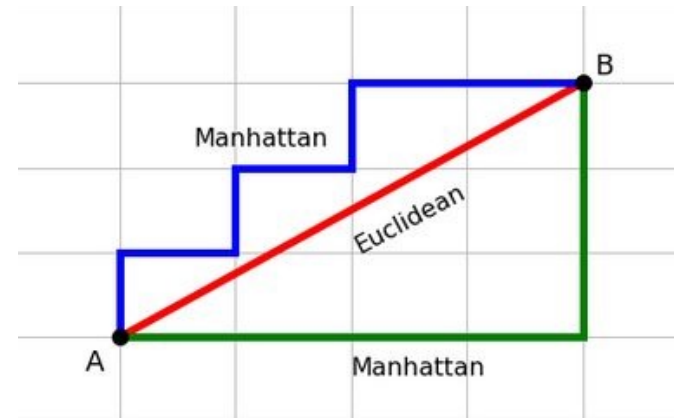
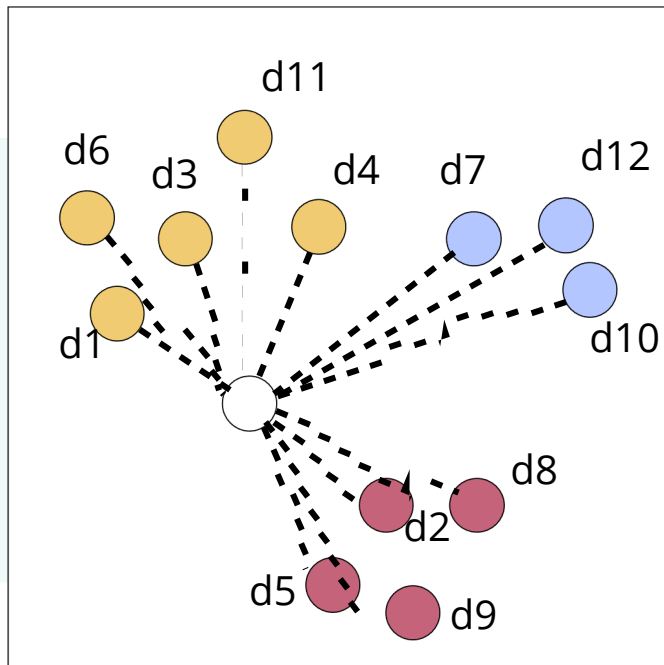


- **Performance measure:** criterion to evaluate a clustering
- **Proximity measure** between data points
 - Similarity: $s(x_i, x_j)$
 - Dissimilarity: $d(x_i, x_j)$



Which clustering is better?
Yellow or green

Proximity measures: distance



Euclidean

$$\sqrt{\sum_{i=1}^k (x_i - y_i)^2}$$

Manhattan

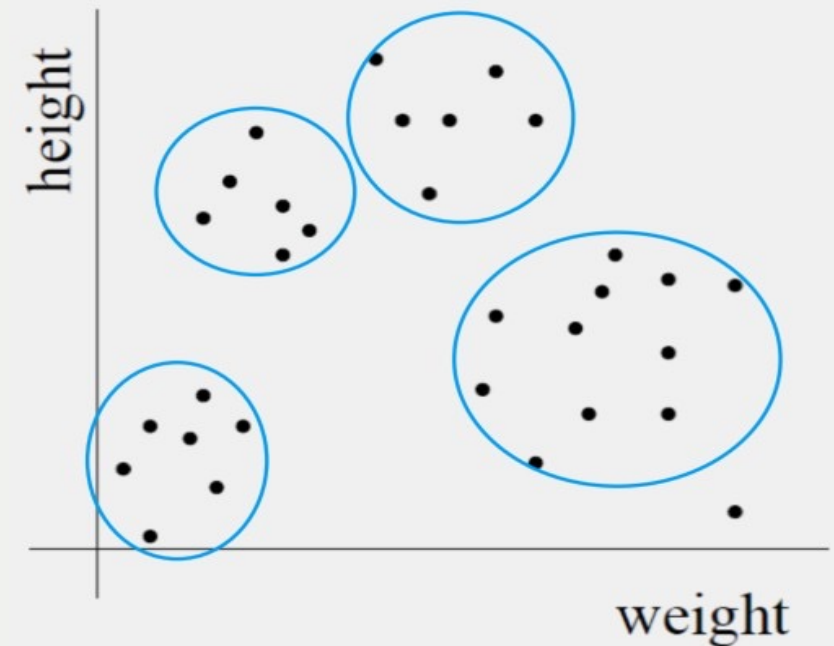
$$\sum_{i=1}^k |x_i - y_i|$$

Minkowski

$$\left(\sum_{i=1}^k (|x_i - y_i|)^q \right)^{1/q}$$

Proximity measures

- **Intra-cluster cohesion (compactness)**
 - Measures how close the datapoints in each cluster are.
 - *For k-means, sum of squared distance from centroid can be used.*
- **Inter-cluster separation (isolation)**
 - Measures distance between clusters
 - *For k-means, the distance between centroids can be used.*



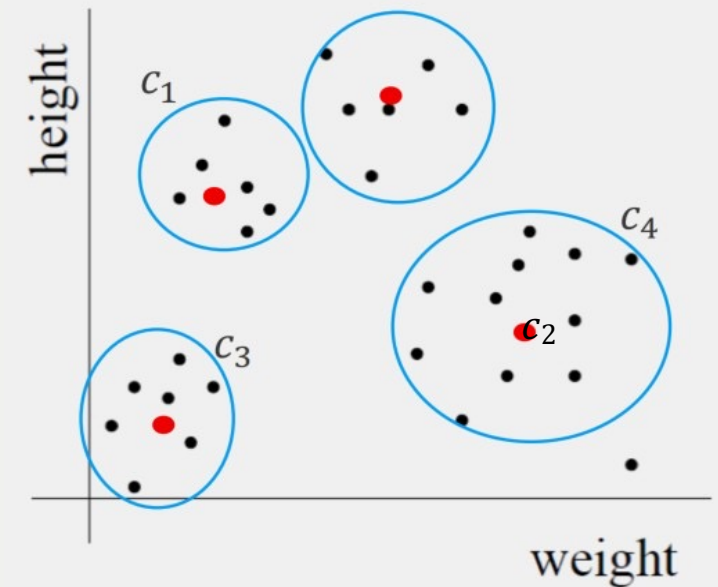
Centroid-based Clustering

Centroid-based Clustering



The **centroid** is the average of all training instances assigned to the cluster

$$\text{centroid} = \frac{1}{|c_k|} \left(\sum_{i \in c_k} x_1^{(i)}, \sum_{i \in c_k} x_2^{(i)}, \dots, \sum_{i \in c_k} x_l^{(i)} \right)$$



Cluster centroid: Example



A two dimensional cluster consists of the following examples:

$(2, 5), (1, 4), (3, 6)$

The centroid is $(2, 5)$

The first attribute's average is $\frac{2+1+3}{3} = 2$

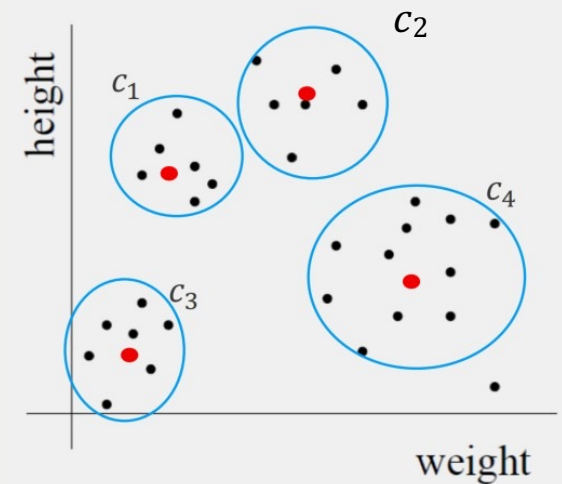
The second attribute's average is $\frac{5+4+6}{3} = 5$

Centroid-based Clustering



- Instances are assigned to clusters based on their Euclidian distance to the cluster centroid:
 - Instances are assigned to the closest cluster

$$d(\mathbf{x}^{(j)}, \mathbf{centroid}) = \sqrt{\sum_l (x_l^{(j)} - centroid_l)^2}$$
$$h(\mathbf{x}^{(j)}) = \min_k d(\mathbf{x}^{(j)}, \mathbf{centroid}_k)$$



Example: Assigning an instance



Suppose that there are 3 clusters with centroids:

$$c_1 = (3, 4), c_2 = (4, 6), \text{ and } c_3 = (5, 7)$$

For instance, $x = (4, 4)$

The Euclidean distances are

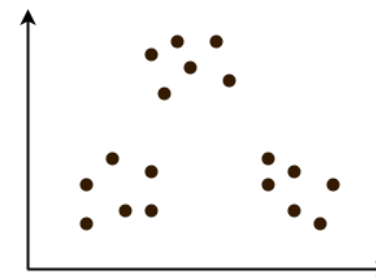
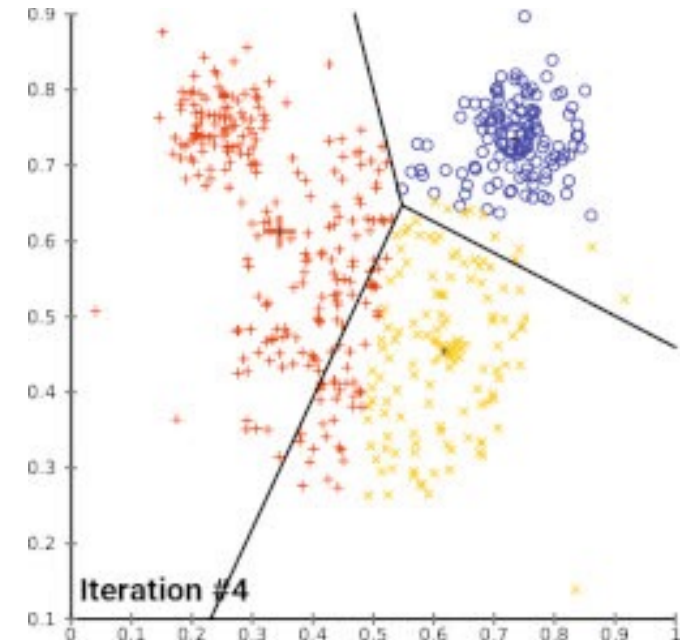
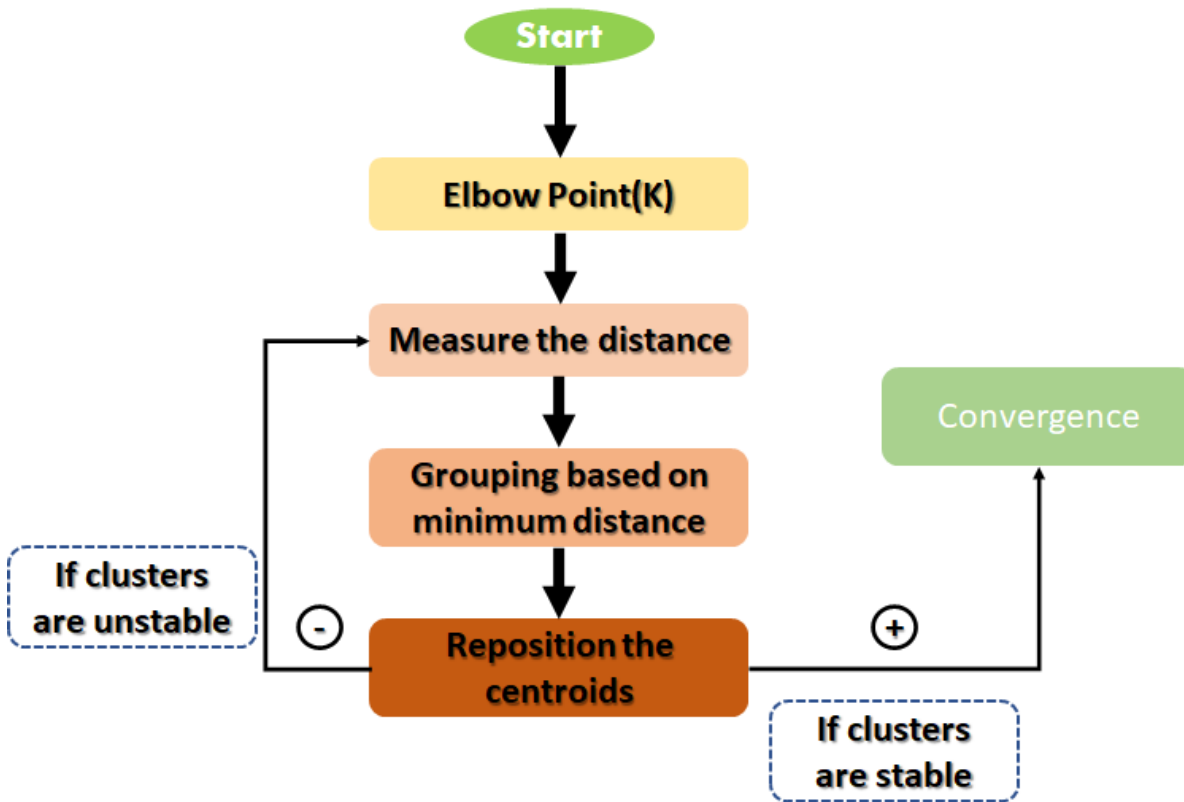
- $d(x, c_1) = 1$
- $d(x, c_2) = 2$
- $d(x, c_3) = \sqrt{10}$

Since $d(x, c_1)$ is the smallest, we conclude that x is assigned to c_1

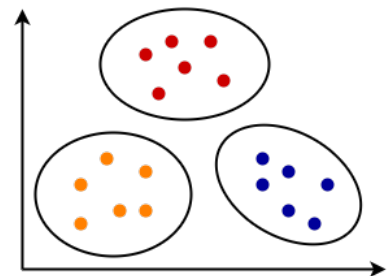
k-Means Clustering

K-Means Clustering

based on the distance

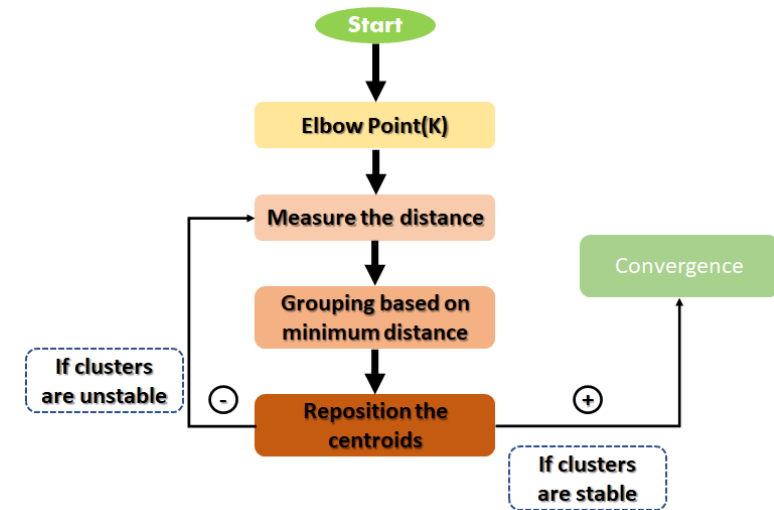
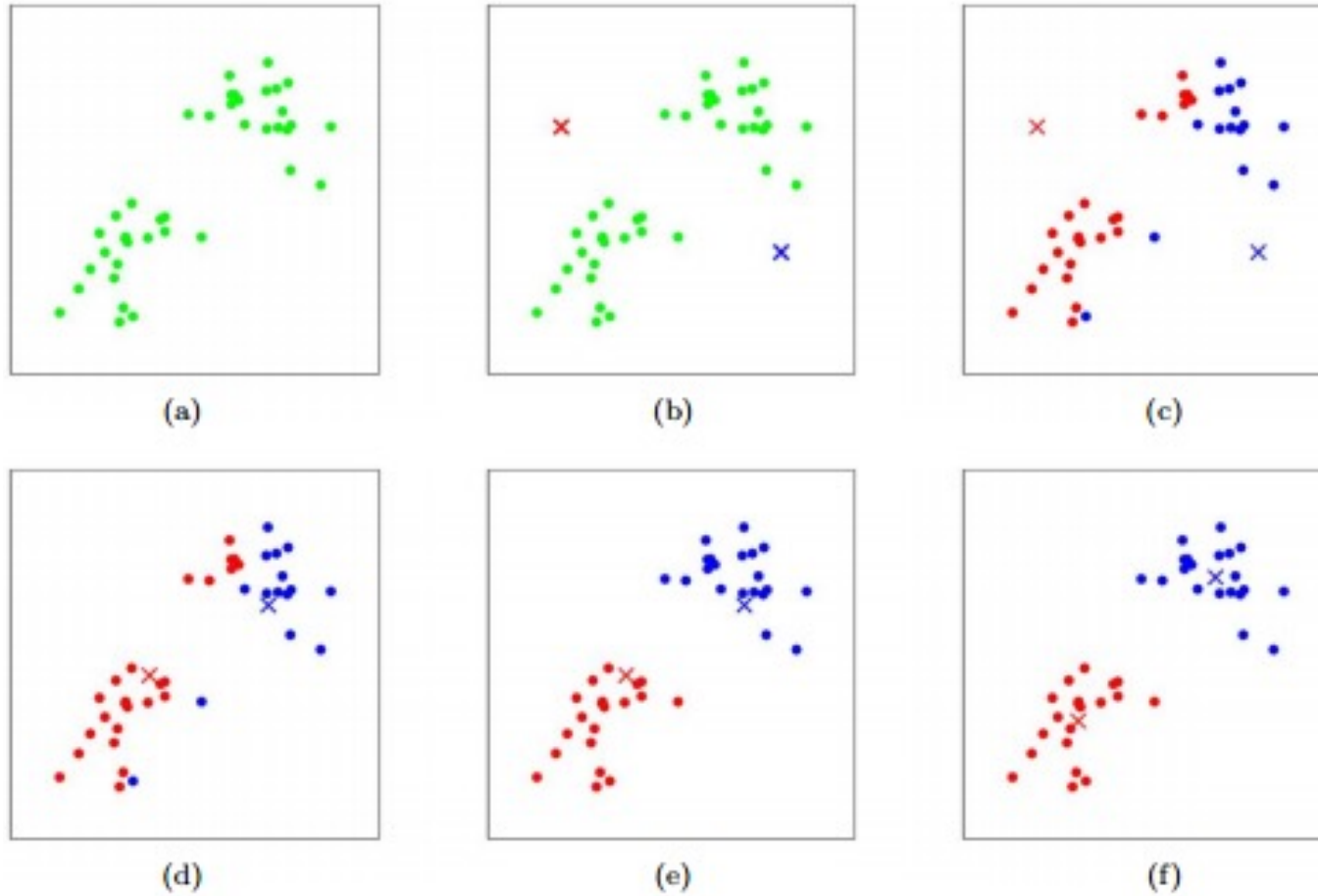


Before K-Means



After K-Means

K-Means Clustering



based on the distance

***k*-Means Clustering** (MacQueen, 1967)



***k*-Means clustering** is a common clustering algorithm

Choose k initial centroids.

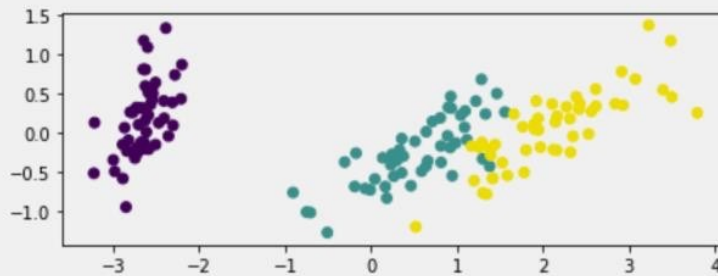
Repeat until convergence

1. Assign each datapoint to the closest centroid.
2. Recompute the centroid using current cluster membership.

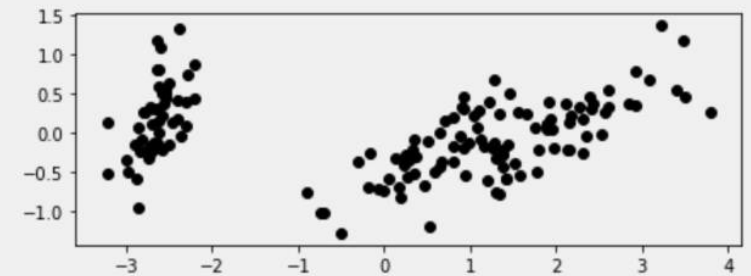
K (the number of clusters to be extracted) is selected by the user.

Tries to minimize summed distance: $SD = \sum_{j=1}^k \sum_{i \in c_j} d(x^{(i)}, \text{centroid}_j)$

k-Means Illustrations

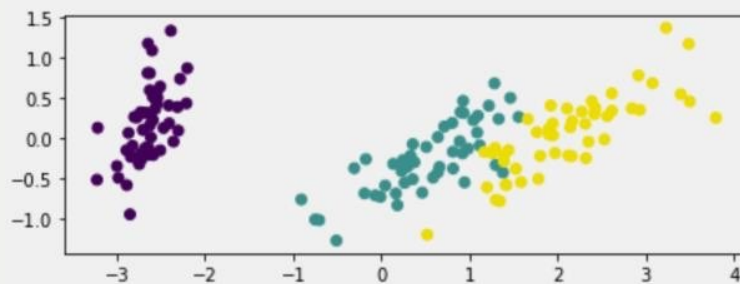


IRIS data converted to 2D using PCA
Color represent the class
(for supervised learning)
(Originally IRIS has 4 attributes, 3 classes)

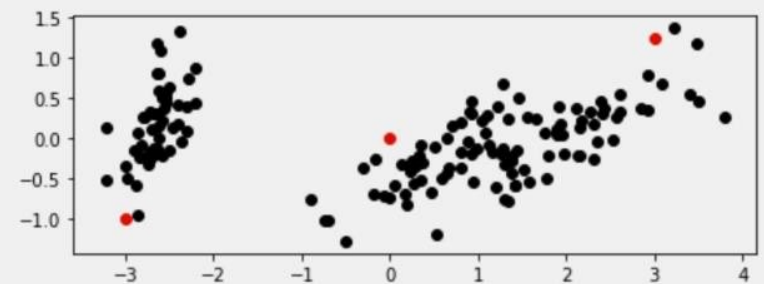


IRIS data converted to 2D using PCA
Input to k-means clustering

k-Means Illustrations

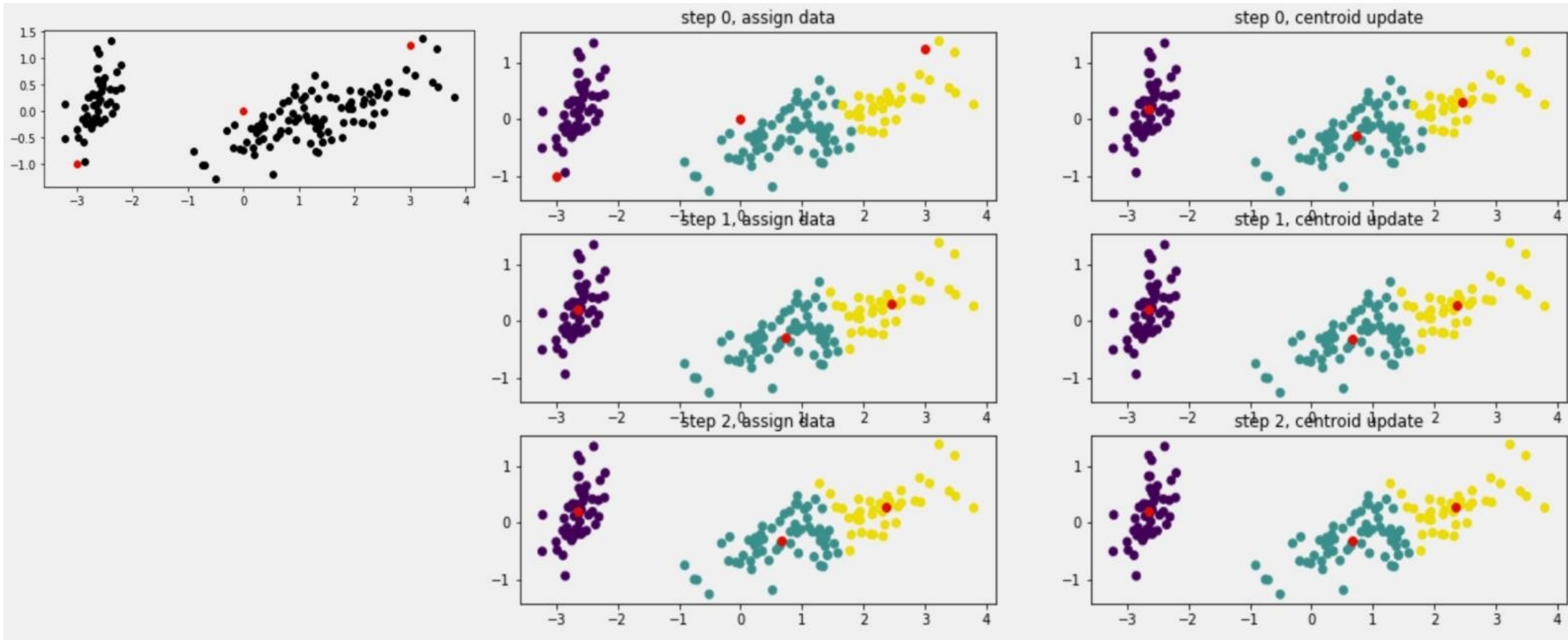


IRIS data converted to 2D using PCA
Color represent the class
(for supervised learning)



Centroid Initialization
K is set to 3

k-Means Illustrations



k-Means Illustrations



Another demo on k-means:

<http://web.stanford.edu/class/ee103/visualizations/kmeans/kmeans.html>

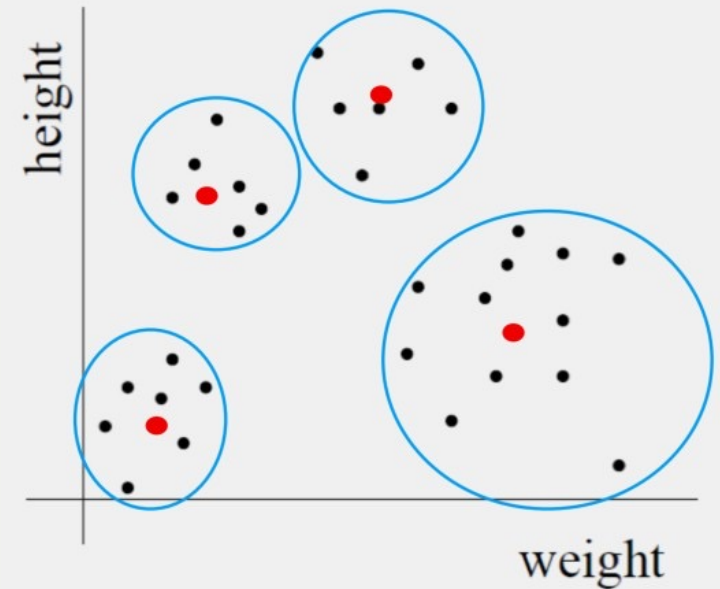
Measuring Cluster Quality



Summed distances (SD)

- It is the sum of all distances of each training instance to its centroid
- This can be thought of as a “loss” measure, and is similar to the RMS error (without the mean/average)

$$SD = \sum_{j=1}^k \sum_{i \in c_j} d(x_i^{(j)}, c_j)$$



Measuring Cluster Quality

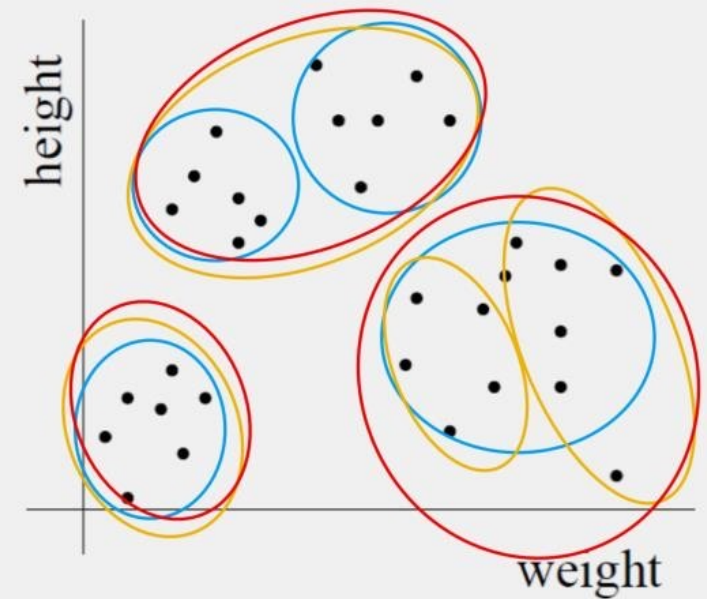


The SD should be minimised:

$$SD = \sum_{j=1}^k \sum_{i \in c_j} d(x_i^{(j)}, c_j)$$

However:

- SD decreases as the number of clusters is increased
- $SD = 0$ if the number of clusters is at least the number of training instances
- SD is **only meaningful when comparing clustering with a similar number of clusters.**



Convergence criteria



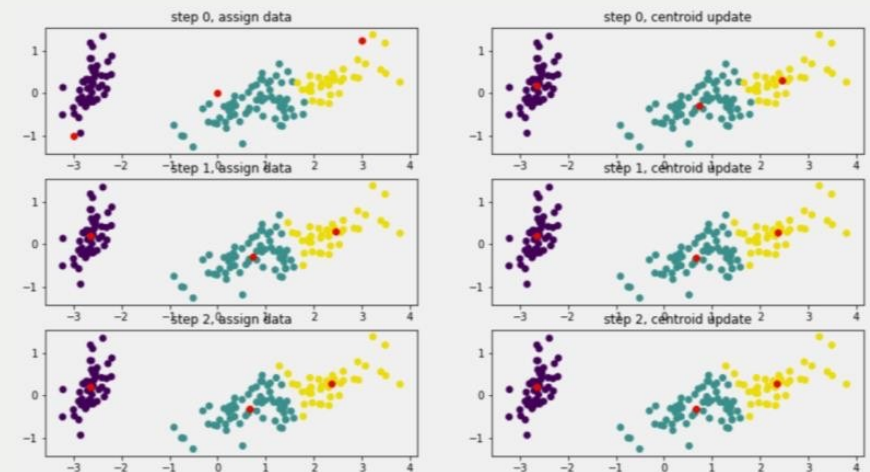
Convergence criterions(Termination)

- No (or minimum) training instances are re-assigned.
- No (or minimal) change in centroid
- Minimum decrease in sum of squared distances

$$SD = \sum_{j=1}^k \sum_{i \in c_j} d(x_i^{(j)}, c_j)$$

- Maximum iterations

Note: Before starting, it is best to **normalise all attribute values**



k-Means strengths



k-Means is the most popular clustering algorithm.

- **Simple:** Easy to understand and implement
- **Efficient:** Time complexity $O(tkn)$ – linear in number of points (n), number of clusters (K) and number of iterations (t).
- For very large datasets there is a **mini batch version of k-means**.

k-Means weaknesses



Will find a **local optimum** if **SD** is used. Usually we can run several k-means with different initialisations to find an approximate global optimal.

The user needs to specify k .

The algorithm is sensitive to **outliers**.

How many clusters?

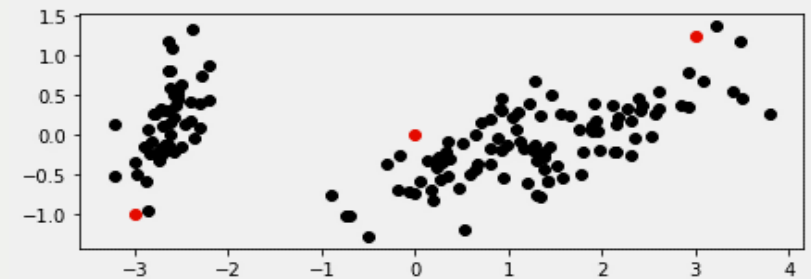


k -means: the number of clusters to be determine prior to clustering

A human observer might be “able to discern” the number of clusters (if the data have low dimensions)

But what about in general?

- In an extreme views:
 - The whole training data set can be viewed as one big cluster, or
 - Each instance as a single-example cluster.
- The “quality” of the clustering can be measured, then select a high-quality k

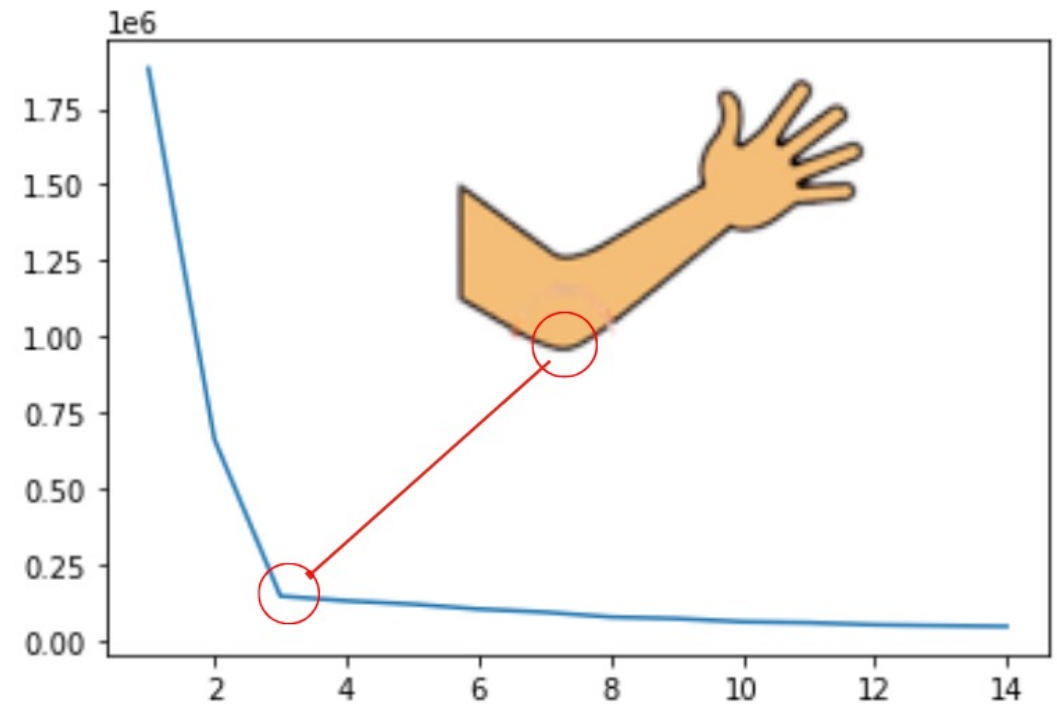


Choosing k?

Elbow method

- Try different values:
 $k = 2, 3, 4, 5, 6, 7$
- Compare performance
(???)

Which metric???



k-Means limitations



The *k*-means algorithm is not suitable for discovering clusters that are not hyper-ellipsoids (or hyper-spheres)



K-means
output



K-means
output



Spectral clustering
output

Hierarchical Clustering

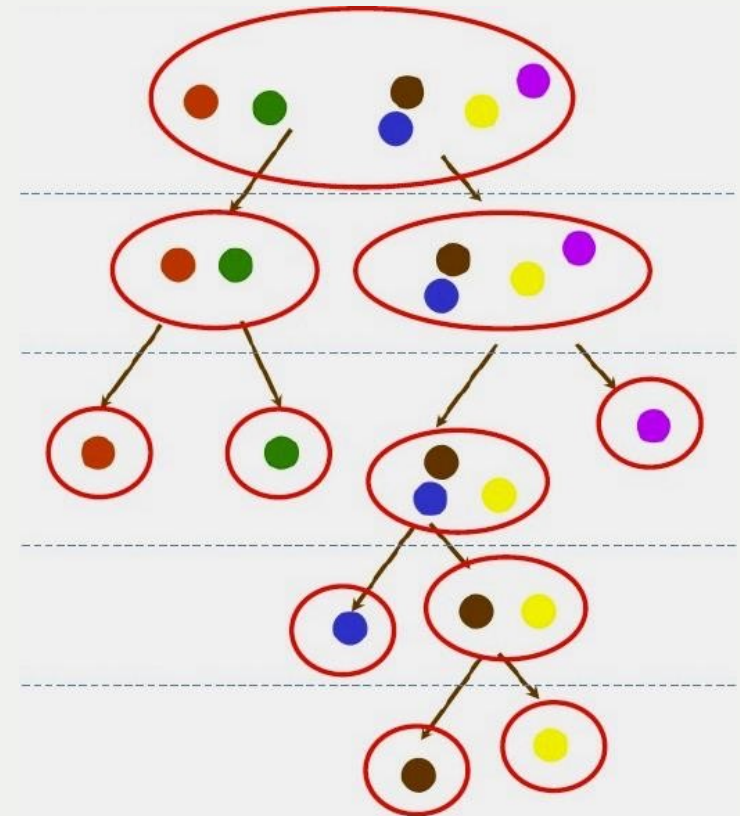
Divisive (top down) clustering



Starts with all data points in one cluster, the root, then

Splits the root into a set of child clusters. Each child cluster is recursively divided further

Stops when only singleton clusters of individual data points remain, i.e., each cluster with only a single point



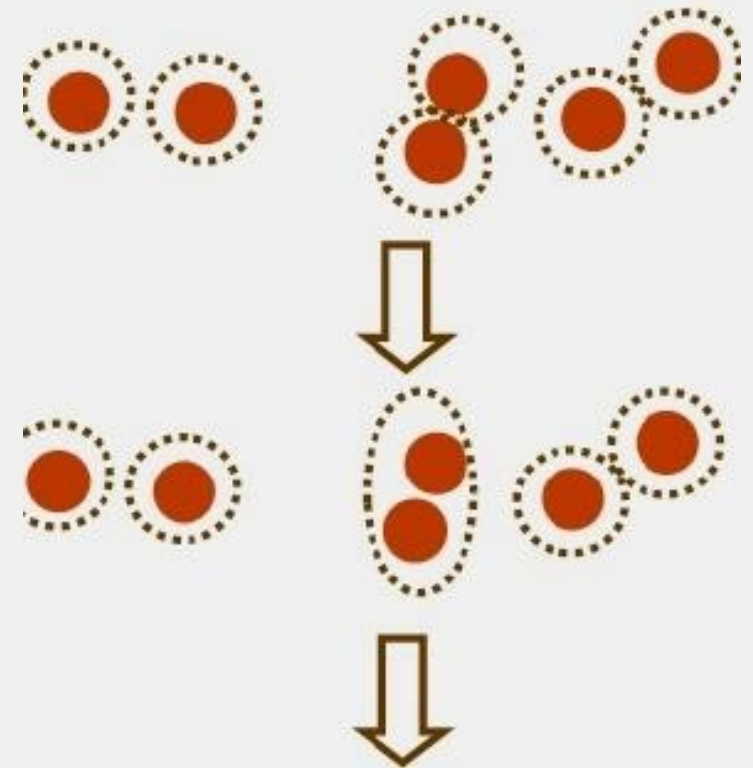
Agglomerative (bottom up) clustering



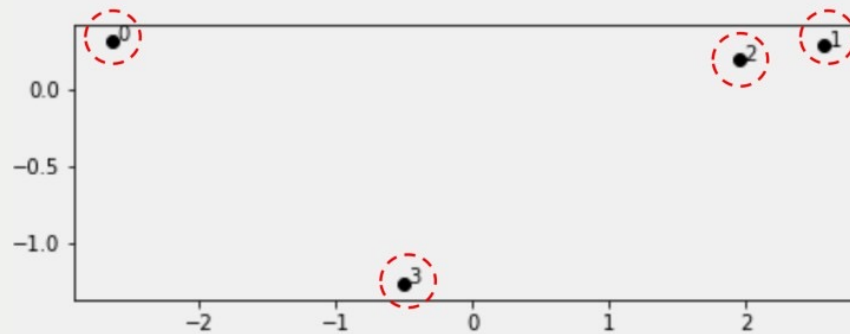
Starts with each data point in individual cluster, then

Merge the **most similar** (or nearest) pair of clusters

Stop when all the data points are merged into a single cluster

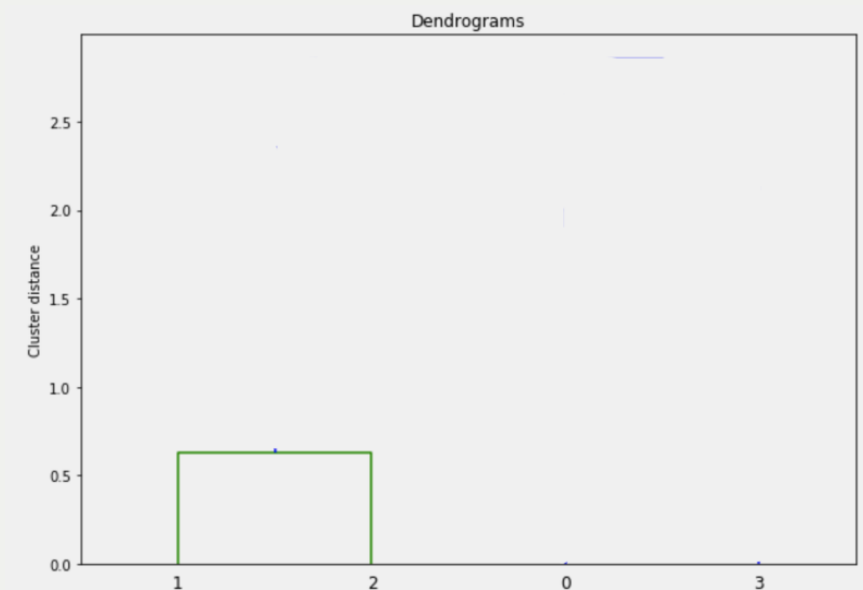
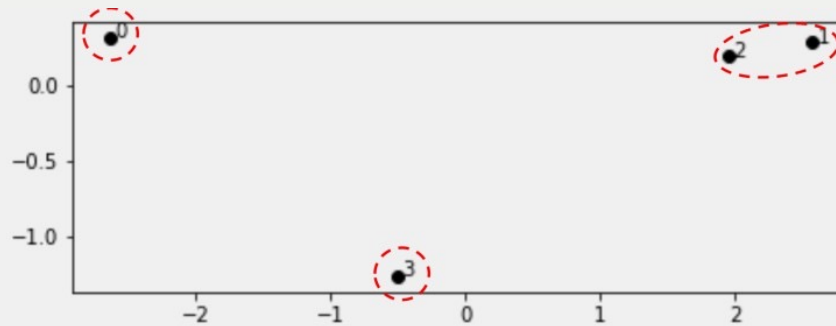


Agglomerative clustering: Illustration

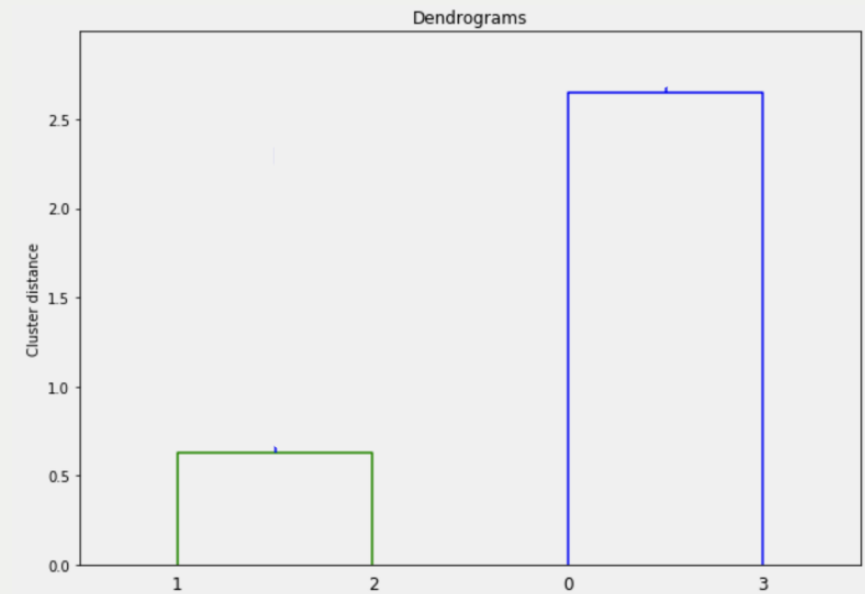
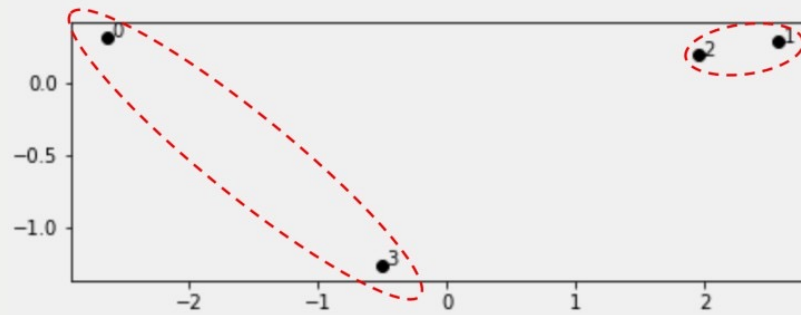


Let's use Minimum distance

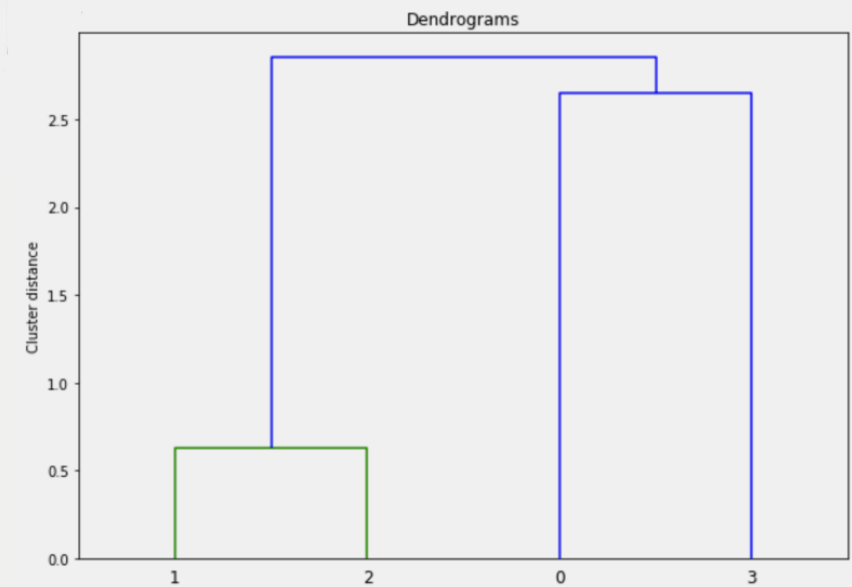
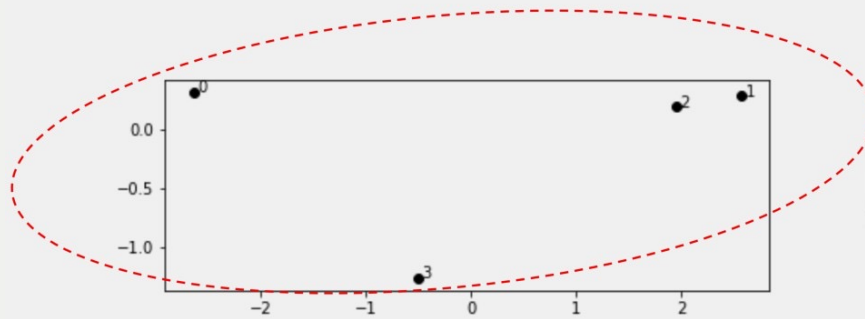
Agglomerative clustering: Illustration



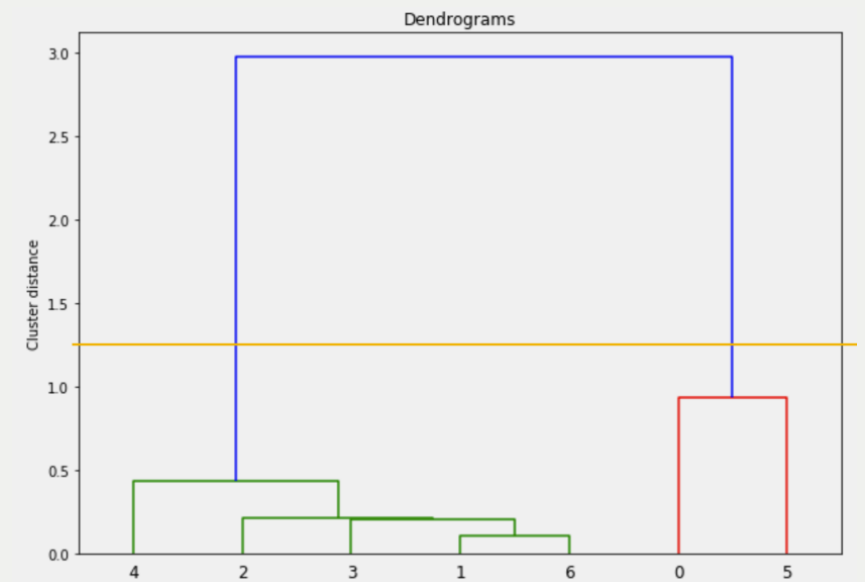
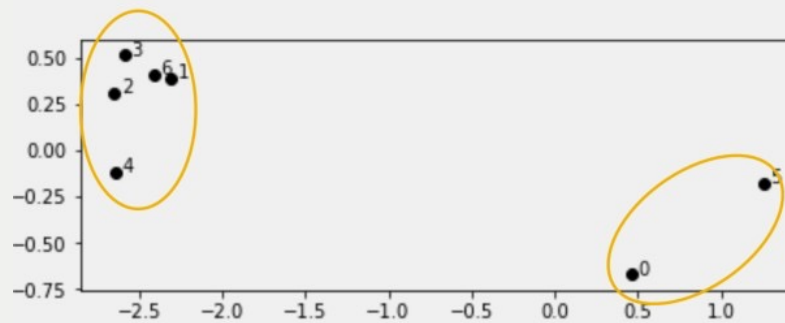
Agglomerative clustering: Illustration



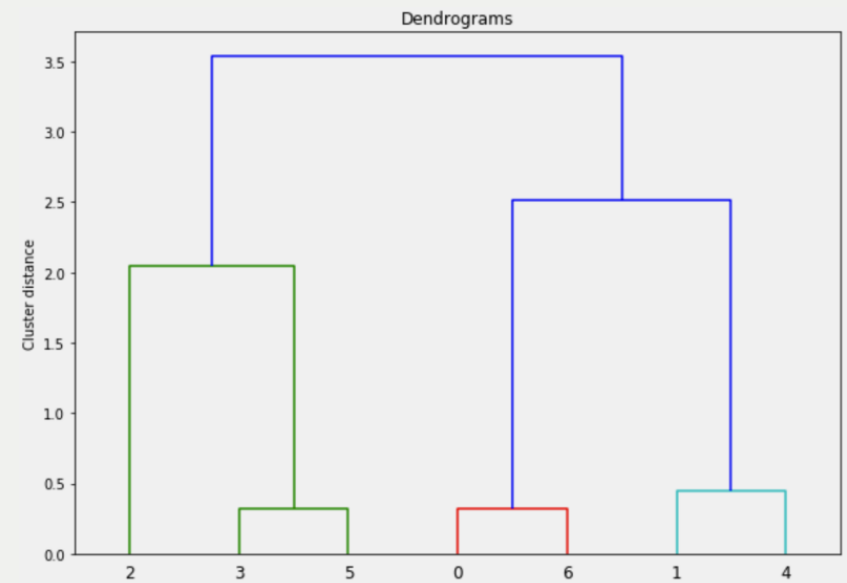
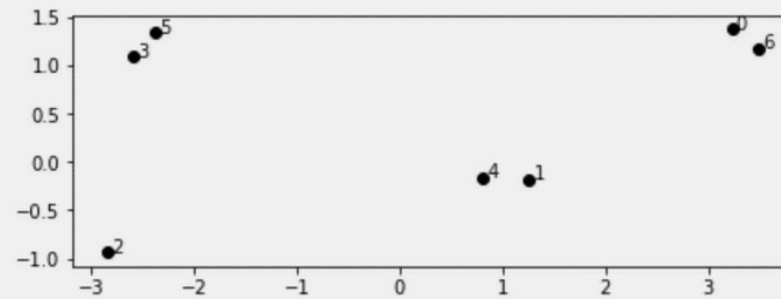
Agglomerative clustering: Illustration



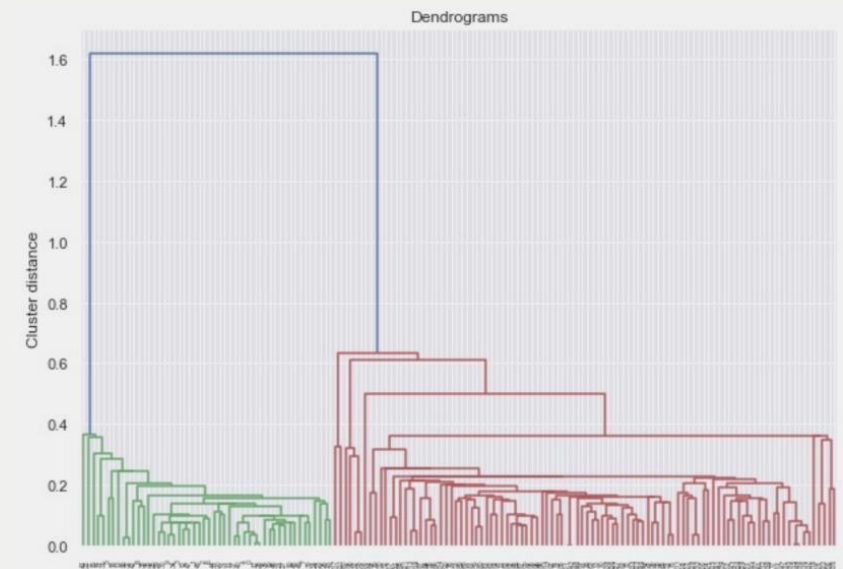
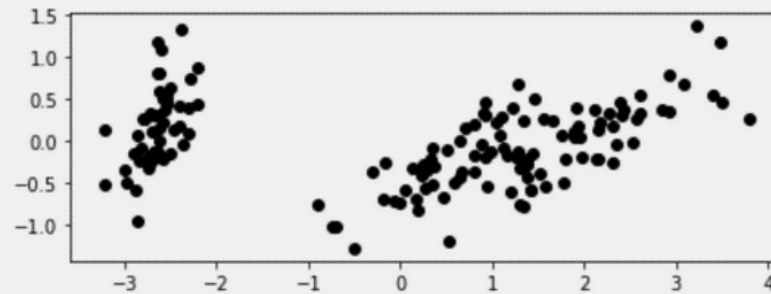
Dendrogram



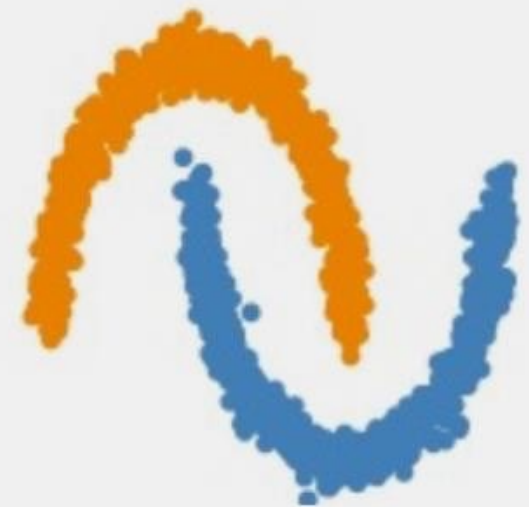
Dendrogram



Dendrogram



Agglomerative



Agglomerative
output

Summary



Clustering

- Approaches
- k -means clustering
- Hierarchical clustering

Thank you!